

Summary of Revisions

Manuscript: A Reproducible Benchmark of Relational Database Access-Layer Performance in Express.js: A Configuration-Specific Comparison on PostgreSQL and MySQL **Author:** Mateusz Miotk (University of Gdańsk) **Journal:** Information and Software Technology

I thank the reviewer for a thorough, multi-part reading. This revision addresses five successive sets of comments — on methodology and the scope of claims, on the CPU and statistical reporting, on the conclusions and language, on presentation, and on the paper's length. A full point-by-point reply, each with a quotation of the revised text, is in the separate Response to Reviewers; the essential changes are summarized below. The main revision changed no measurement — it is one of framing, scoping, and length; the deeper follow-up points noted at the end add clearly-scoped *supplementary* measurements that leave the primary matrix untouched.

Scope of claims and methodology

- The estimand is renamed from "default-configuration" to ****documentation-selected**

implementation-and-strategy**: choosing the first documented API is a reproducible selection heuristic, not the library's actual default configuration.

- The comparison is now read at **three explicit, separated levels**: the primary

implementation-and-strategy comparison, the same-SQL raw-path **bounding** control, and the individual mechanisms, which change together and are **not** causally isolated.

- The operating conditions are scoped: the full five-pattern, two-engine matrix is the fixed

high-load point, while the equal-utilization and equal-CPU conditions are a limited set restricted to the deep fetch. "capacity", "saturating throughput", and "overload" are reserved for the deep-fetch sweep, where a throughput knee was actually measured.

- A new **experiment-scope table** (Supplement Table S32) tabulates every experiment's

patterns, engines, layer count, and repeated runs.

CPU and statistics

- The equal-CPU experiment is **demoted to an exploratory sensitivity check**; the earlier

"at most a few percent" wording is corrected (the ORM per-core values wobble 10–17% run-to-run with no monotonic gain from more cores).

- The bootstrap intervals are stated to capture **within-campaign**, run-to-run variability

on this host, not variation across machines or versions; adjacent-layer p99 gaps of one or two milliseconds sit at the millisecond measurement floor and are not read as real.

- The raw per-replicate MySQL-insert distributions (Supplement Figure S1) are brought into

the Results body.

Conclusions and language

- "access-layer overhead" wording is limited where it implied an isolated measurement (the

primary comparison bundles library, query strategy, round-trips, and mapping); the "thin layers are a safer default" recommendation becomes **"benchmark the relation-heavy hot path for**

the specific application"; an explicit **configuration-specific** label is added to the Conclusion; and novelty is scoped throughout to "the sources searched".

Presentation, length, and consistency

- Rather than split the paper (which would read as salami-slicing, since both threads share one research question), the main text is **condensed by about 24% (51 → 43 pages; body 11,356 → 8,669 words)** so the central message is not lost among the controls and sensitivity analyses. Detail is **moved to the supplement, not deleted**: new *Statistical Methods*, *Measurement Details*, and *Benchmarking-Pitfalls Checklist* sections. The same-SQL control, the RQ1–RQ3 answers, and Threats to Validity remain prominent in the main text. The manuscript is 12,216 words, well under the 15,000-word IST limit.

- A pre-submission consistency sweep confirmed that the pinned library versions match the lockfile, the DOI and repository link resolve, and every headline number matches its table; it corrected a coefficient-of-variation figure ("within 4.2%", not 4.0%, on the reads), removed redundant per-cell median confidence intervals from the paired significance table that had disagreed with the pattern table, and reconciled two loose prose phrasings.

Deeper follow-up points (6.1–6.6)

Six deeper follow-up points were addressed after the main revision and released incrementally:

- **6.1** — the same-SQL result is renamed a **standardized (same-SQL) contrast**, not a "bound":

absent interaction assumptions, the raw-path spread does not bound the intrinsic library effect.

- **6.2** — the semantic-equivalence gate is **strengthened to validate write state** (a new `bench/verify-writes.mjs` confirms exact field values, row-count changes, and transactional rollback through an independent native-driver connection), not merely a 2xx status.

- **6.3** — the **comparability protocol is specified normatively** and independently of the case

study, in a new Methodology subsection (inputs; mandatory stages; pass/fail cell admission; output interpretation; applicability limits).

- **6.4** — RQ1 is narrowed to the performance of each layer's ****documentation-selected** implementation strategy, **and a second** performance-conscious co-primary regime****** — the faster of a layer's two documented deep-fetch loading strategies, admitted only where byte-identical to the documentation-primary output — is measured on one harness and reported (new Supplement Table S34). The finding reinforces rather than overturns the documentation-primary reading: the documented strategy is already the faster one for most layers, and where it is not (Objection on MySQL) the performance-conscious ORM deep fetch still sits below the native driver.

- **6.5** — **capacity is now characterized for all five patterns**, not only the deep fetch. A per-pattern concurrency sweep (connection ladder 1–200, every layer, both engines; new Supplement Table S35) locates each pattern's throughput knee at or below 50 connections, so the fixed 50-connection operating point places all five patterns at high utilization of their own capacity

— making the cross-pattern p99/spread comparison one at comparable, measured relative utilization rather than unknown fractions of capacity. The equal-utilization open-loop tail and the exploratory equal-compute check remain demonstrated on the deep fetch, and the manuscript now labels those as deep-fetch-only.

- **6.6** — the over-strong claim that four controls "isolate the access layer as the only variable"

is **corrected** (prose only). The controls hold the shared conditions fixed (pool, logical task, physical state, observer) so the *configured access-layer implementation bundle* is the only deliberately varied factor; they do not isolate the library alone, since the bundled differences (SQL, query count, protocol, loading strategy, hydration) are the treatment, not confounds. Two related-work "isolates" usages were reworded to "treats"/"varies"; the isolation *disclaimers* elsewhere were verified correct.

Points 6.2, 6.4, and 6.5 add new, clearly-scoped *supplementary* measurements (a write-state oracle, a deep-fetch regime, and a per-pattern capacity sweep); point 6.6 is a wording correction only. The primary measurement matrix and every previously reported primary number are unchanged.

Point 7 (minor concerns)

Six further wording/labeling corrections, no measurement changed: the abstract's version-specific "Prisma 7 sits mid-pack" clause was removed (de-loading it); "the most widely used" Node.js framework was softened and temporally qualified to "has long been among the most widely used"; Table 1's "Independent?" column was relabelled the neutral **"Vendor involvement"** (a source/conflict-of-interest property, not a methodological one); the threats claim that a hot-cache regime "maximizes layer differences" became "expected to make application/access-layer overhead more visible"; the primary text now discloses the one Drizzle/MySQL-insert cell with 24 rather than 25 runs; and "independent runs" was reworded to "repeated runs" in the captions and REPRODUCE (all repetitions share a host and campaign).

Point 8 (methodological and statistical assessment)

The reviewer's assessment was favourable ("substantially above the level of many benchmark studies ... no clear statistical error requiring rejection"). Five precision/scoping fixes, no measurement changed: (1) Methodology now states explicitly that *"p99" throughout is the median run-level p99* (median across replicates of each run's p99), not the p99 of the pooled request distribution*; (2) **the within-campaign scope of the bootstrap intervals is reinforced where they are used, so they are not read as deployment-general**; (3) **the coarse millisecond p99 resolution is noted not to finely separate near-adjacent layers**; (4) **following the reviewer's suggestion to foreground effect sizes, the paired-significance table (former main-text Table 7) was moved to the supplement (new Table S36)*, the main text now leading with paired geometric-mean ratios and per-replicate dominance and noting the post-hoc tests' inference is conditional on the ranking that selects the pairs**; (5) *the secondary experiments on layer subsets or few replicates (Supplement Table S32) are labelled sensitivity evidence** on the tested subset, not claims about all eleven implementations. Moving Table 7 drops the main-text float count from eight to seven.

Point 9 (reproducibility and artifact assessment)

The reviewer rated this "one of the manuscript's strongest dimensions." I verified the five-item pre-submission checklist — all satisfied: (1) the Zenodo tarball byte-matches `git archive` of

the tag (identical sha256); (2) every statistic traces to archived raw data (`MANIFEST.md`, 35 files), the one disclosed exception being the deterministic S2 query-count table; (3) `environment.txt` and `db-config.md` capture kernel/runtime, CPU/NUMA topology, governor/turbo, virtualization, lockfile hash, and database configuration; (4) tables rebuild from raw data with pinned dependencies and no network; (5) an explicit dual license (MIT code, CC BY 4.0 paper/data). For the AI-declaration policy point, the declaration was **split** into the Elsevier-standard writing declaration and a separate *AI-assisted research software* statement (author responsibility; the AI-assisted analytical code passes 19/19 estimator unit tests). No body change; the AI declarations are excluded metadata.

The replication package for this revision is archived at Zenodo, DOI 10.5281/zenodo.21472130 (release v1.7.4). Every table reproduces from the archived raw data, whose checksum manifest verifies 35/35.